

SYSTEM ANALYSIS AND DESIGN

WEB-BASED ISSP MANAGEMENT SYSTEM

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I. Introduction

1.1 Purpose

The Web-Based ISSP Management System is designed to streamline the preparation, management, consolidation, and monitoring of Information Systems Strategic Plan (ISSP) data within the agency. The system aims to provide a centralized platform for encoding ICT projects, managing budget requirements, monitoring project implementation, and generating consolidated ISSP reports aligned with the Revised ISSP Template 2026.

1.2 Project Overview

The system is a web-based application that allows authorized personnel to manage ICT-related submissions and reports through a centralized and role-based platform.

The system supports:

- ICT project submission
- Budget management
- Project consolidation
- Report generation
- Submission tracking

II. Tracking & Reporting

Key Milestones & Required Events (May 26, 2026 - July 16, 2026)

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PHASE	MAJOR MILESTONE/EVENT	START DATE	END DATE	DURATION
I. System Analysis and Planning	Requirements gathering, workflow analysis, architecture planning, and database design	May 26, 2026	May 28, 2026	1 Week
II. UI/UX Design and Authentication Development	UI wireframes, dashboard design, login system, role management, and database setup	June 1, 2026	June 4, 2026	1 Week
III. Core Module Development	ICT project management, budget management, and submission workflow	June 8, 2026	June 18, 2026	2 Weeks
IV. Approval and Consolidation Workflow	Approval process, validation, consolidation module, and monitoring features	June 22, 2026	July 2, 2026	2 Week
V. Reports and Dashboard Development	Dashboard analytics, PDF reports, and printable ISSP reports	July 6, 2026	July 9, 2026	1 Week
VI. Testing and Deployment	Integration testing, bug fixing, user acceptance testing, revisions, user manual, and deployment preparation	July 13, 2026	July 16, 2026	1 Week

Weekly Sprint Tracking

Week	Date (Monday Check-in)	Sprint Goal	Key Tasks	Status
Sprint 1	May 26 – May 28, 2026	System Planning and Analysis	Requirements gathering, workflow analysis, database planning, and system architecture design	Done
Sprint 2	June 1 – June 4, 2026	UI/UX and Authentication Setup	UI wireframes, dashboard design, login system, role management, and database creation.	Ongoing
Sprint 3	June 8 – June 11, 2026	ICT Project Module Development	Project encoding, submission forms, editing, and project status tracking	Pending

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Week	Date (Monday Check-in)	Sprint Goal	Key Tasks	Status
Sprint 4	June 15 – June 18, 2026	Budget and Submission Workflow	Budget management, submission workflow, and monitoring dashboard	Pending
Sprint 5	June 22 – June 25, 2026	Approval Workflow Development	Department Head approval, validation process, remarks, and notifications	Pending
Sprint 6	June 29 – July 2, 2026	Consolidation and Monitoring Module	Project consolidation, monitoring, and centralized tracking	Pending
Sprint 7	July 6 – July 9, 2026	Reports and Export Features	PDF reports, Excel export, printable reports, and dashboards	Pending
Sprint 8	July 13 – July 16, 2026	Final Testing and Deployment	Integration testing, bug fixing, user acceptance testing, revisions, and deployment preparation	Pending

III. Technical Approach and Specifications

4.1 Architecture

The system will utilize a Three-Tier Web Architecture composed of the Presentation Layer, Application Layer, and Database Layer to ensure organized communication, maintainability, and centralized management of ISSP-related data.

The Presentation Layer handles the user interface including dashboards, forms, monitoring pages, and reports. The Application Layer manages the core system functionalities such as authentication, ICT project management, approval workflow, consolidation, and report generation. The Database Layer stores centralized records including users, ICT projects, budgets, monitoring data, and audit logs.

The architecture supports role-based access control for Employees, Department Heads, and Administrators.

LAYER	COMPONENT	DESCRIPTION
User Layer	Administrator	Manages users, system

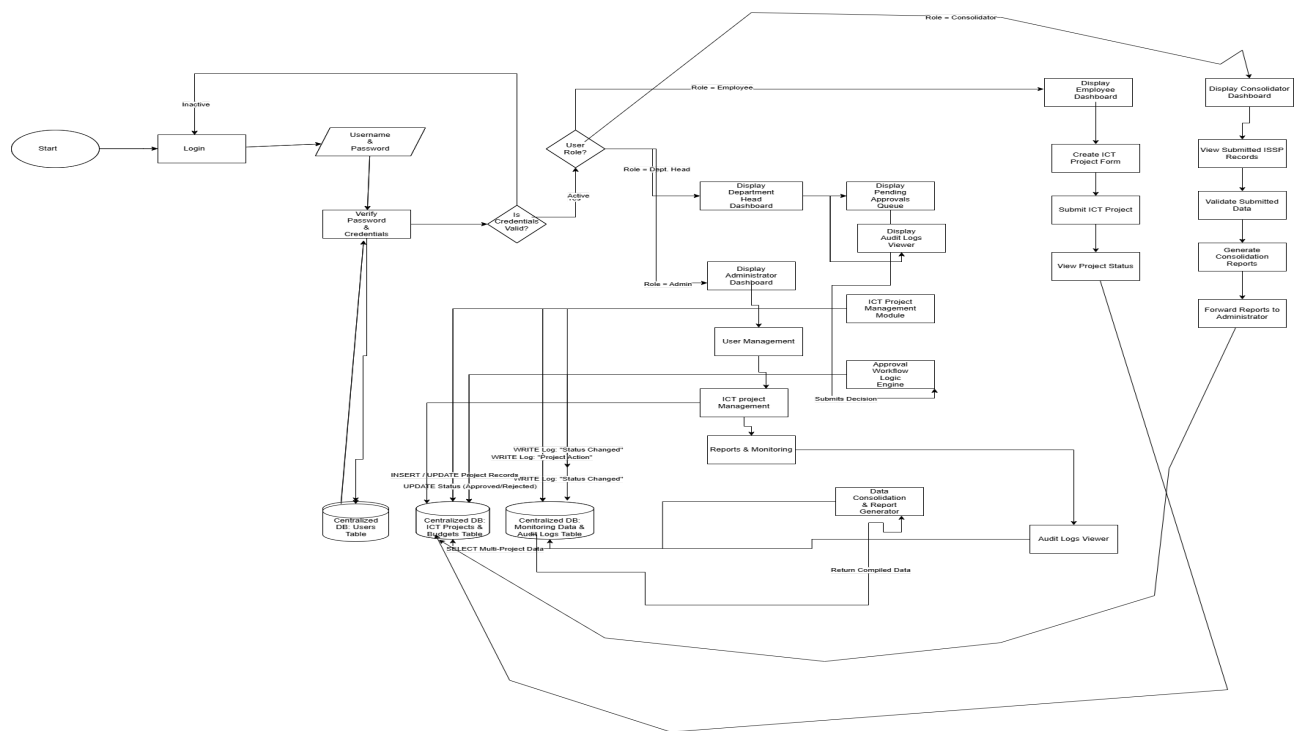
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		configurations, and monitoring
	Department Head	Reviews, approves, and monitors ISSP submissions
	Consolidator	View all departmental submissions, consolidate data, generate consolidated reports.
	Employee	Encodes and submits ISSP records
Presentation Layer	Bootstrap 5 UI	Provides responsive design and user interface components
	DataTables	Supports searching, sorting, filtering, and pagination
Application Layer	PHP 8	Server-side scripting language
	CodeIgniter Framework	MVC framework for system development
	Authentication Module	Handles login, logout, password security, and session management
	User Management Module	Manages user accounts, roles, departments, and positions
	ISSP Management Module	Manages ISSP creation, updates, and tracking
	Consolidation Module	Allows Consolidators to collect, review, validate, and consolidate ISSP submissions from different departments.
	Workflow Module	Manages Draft, Pending, Approved, Rejected, and Revision statuses
	Notification Module	Sends system alerts and

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		status updates
	Attachment Management Module	Handles file uploads and document storage
	PDF Report Generation Module	Generates downloadable ISSP reports automatically
	Logging Module	Tracks system and user activities
Database Layer	MySQL Database	Stores all system data

4.2 Flowchart



Flowchart GDrive Link: [📁 ISSP_Flowchart](#)

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4.2 Technology Stack

CATEGORY	TECHNOLOGY/TOOL	JUSTIFICATION
Frontend	Bootstrap 5, HTML, CSS, JavaScript	Responsive and user-friendly web interface
Backend	PHP 8, CodeIgniter 4	Structured and modular web development
Database	MySQL	Centralized relational database management
DevOps Environment	/ Docker, Docker Compose	Provides consistent development environment, isolates services (Apache, MySQL, phpMyAdmin), and ensures easy deployment across different machines
Development	VS Code, Git, GitHub	Standard tools for version control and coding